

Pibo

Setup & Administration

Robots, routers and the classroom network

Who this is for

The person who installs the robots and looks after them — not the teacher running the lesson. Teachers should use the *Pibo Teacher's Guide*.

Contents

1	How Pibo joins a network
2	The router supplied with the robots
3	Installing in a classroom
4	Preparing tablets and laptops
5	Checking a robot
6	Connecting to a different network
7	Factory reset
8	Troubleshooting
A	Appendix A — Defaults and addresses

The short version

The robots and the router are delivered already matched to each other. Plug the router in, switch the robots on, and they connect by themselves. Sections 1 and 2 explain why, and what not to change.

One router per room, ten robots

Use a single router in each room, and plan for ten robots on it. Never put two routers with the robot network in the same room — see section 3.

1 How Pibo joins a network

Every robot leaves the factory set to join a 5 GHz network called `pibo`, and the router supplied with the robots is already configured to provide exactly that. Nothing has to be matched up on site.

At every start-up

1. Pibo looks for the Wi-Fi network stored in its settings — `pibo`.

↓ found

2a. It joins, receives an address, and shows that address on the chest screen. Ready to use.

↓ not found

2b. It starts a network of its own named `pibo-` plus the last eight characters of its serial number — the same **SN** shown on the chest screen. This is called **AP mode**.

The check repeats every ten seconds for as long as the robot is on. If the router is switched on later, or comes back after a power cut, the robots join by themselves and shut their own networks down. They do not need restarting.

The chest screen

```
SN: cd488e95
I P: 192.168.114.138
AP: pibo
```

Connected — open the **I P** address in a browser

```
SN: cd488e95
I P:
AP:
```

AP mode — the router was not found

SN is the robot's number, **I P** the address to open in a browser, and **AP** the network it joined. An empty **I P** line means AP mode. The **SN** also tells you which `pibo-...` network belongs to which robot, which matters when twenty of them are on at once.

AP mode means there is no internet

A robot running its own network has no route to the outside world — it only has one radio, and it is busy being an access point. This is expected behaviour, not a fault.

2 The router supplied with the robots

A TP-Link Archer AX53 is supplied, already configured. Plug in power, and if the school has internet, plug the school's cable into the router's **WAN** (internet) port. Nothing else is needed.

What is already set

Setting	Value
5 GHz network name	pibo — the robots join this one
2.4 GHz network name	pibo-24 — for tablets and older laptops
Password (both)	!pibo0314
5 GHz channel	36, fixed
Channel width	80 MHz
Wireless mode	11ac (Wi-Fi 6 / 11ax turned off)
Security	WPA2-PSK (AES)
Band steering (Smart Connect)	Off, so the two names stay separate
Automatic firmware update	Off

Do not change the wireless settings

Each value above is there for a reason, and several of them will stop the robots connecting if changed. In particular: the channel must stay fixed rather than *Auto*, the mode must stay 11ac, and the security must stay WPA2-PSK.

Leave automatic firmware update switched off as well. An update can put the channel back to *Auto*, which takes every robot in the room off the network at the same time.

How many robots per router

One router per room, and plan for ten robots on it. The hardware can carry more, but ten keeps the classroom comfortable: pictures from the camera keep flowing, and a room full of children pressing **Run** at the same moment does not cause a queue.

Adding a second router to the same room does not help, and makes things worse — see section 3.

Internet is optional

Speech, vision, face recognition and the language model all run on the robot itself. If the school has no internet, or its internet fails, the lesson is unaffected. Connect the school's cable to the **WAN** port when it is available — it is useful for the teacher's tablet, not for the robots.

3 Installing in a classroom

Ten robots, one router

1. Place the router where it can be seen from most of the room, not inside a cupboard or behind a metal cabinet.
2. Connect power. If the school has internet, connect its cable to the **WAN** port.
3. Wait about a minute, then check on a tablet that the networks `pibo` and `pibo-24` are visible.
4. Switch the robots on, a few at a time.
5. After about a minute each chest screen shows an **IP** address. Write down the SN and IP of each robot and give the list to the teacher.
6. Connect the tablets to `pibo-24` (or to `pibo` if they support 5 GHz).

Never use two robot routers in one room

Two routers with the same network name make the robots flap

Both would broadcast `pibo`. Each robot then hears two of them, keeps deciding the other one looks better, and moves back and forth all lesson — dropping its connection every time. This is worse than having too many robots on a single router, and it is hard to diagnose because everything looks correctly configured.

One room, one router. Do not add a second, and do not bring a spare router into a room where one is already running.

More than ten robots at a school

Split them between rooms, one router in each, and keep the rooms far enough apart that a robot in one room cannot hear the other room's router. A wall or two of separation is usually enough; check by standing in one room with a tablet and looking at how strong the other `pibo` network appears — if it does not show up at all, the separation is sufficient.

If the robots must all be in one room, keep them on the single router and expect the camera examples to feel slower when every child runs one at the same moment. Having half the class work on non-camera examples spreads the load.

Where to put the robots

Keep robots off the floor and away from metal shelving, and leave a clear space around each one so it can move its arms and legs without hitting anything. Line of sight to the router gives the best signal.

4 Preparing tablets and laptops

The `pibo` network usually has no internet behind it. Windows, Android and iPadOS all treat a network with no internet as a poor one and move away from it on their own — which looks, to the teacher, like the robots disconnecting. A few minutes per device prevents this for good.

Why it happens

Each of these systems checks whether a Wi-Fi network can actually reach the internet. When it cannot, and another remembered network is within range, the device switches to that one instead, usually without saying anything. The robots are still there; it is the tablet that left.

The simplest fix: give the router internet

If a school network cable can reach the classroom, connect it to the router's **WAN** port. The `pibo` network then has internet like any other, nothing tries to leave it, and the rest of this section becomes unnecessary. Try this first.

Windows

Windows rejoins any remembered network when it comes into range. Stop it joining anything but the robot network.

1. Open **Settings** → **Network & internet** → **Wi-Fi** → **Manage known networks**.
2. For **every** network except `pibo` and `pibo-24`, open it and turn **Connect automatically when in range** off. Networks that will never be used again can be removed with **Forget**.
3. Open `pibo` or `pibo-24` and make sure **Connect automatically** is on.
4. If the laptop has mobile broadband or a tethered phone, switch it off during lessons.

“No internet, secured” is normal

Windows labels the robot network this way and puts a warning mark on the Wi-Fi icon. The connection to the robots works perfectly. Tell teachers to expect it — otherwise somebody will try to fix it by switching networks, which is the very thing to avoid.

Android tablets

1. **Settings** → **Network & internet** → **Internet**. Open the gear beside each saved network other than the robot network and turn **Auto-connect** off, or forget it.
2. In the same screen open **Network preferences** and turn off **Switch to mobile data automatically**. The wording varies slightly between manufacturers.
3. If the tablet has a SIM, simply turning mobile data off during the lesson is just as effective.
4. Android may ask *This network has no internet access. Stay connected?* — answer yes and tick **Don't ask again for this network**.

iPad

1. **Settings** → **Wi-Fi**. Tap the ⓘ beside each other network and turn **Auto-Join** off.
2. Tap ⓘ beside `pibo` and make sure **Auto-Join** is on.

3. On models with mobile data, turn **Settings** → **Mobile Data** → **Wi-Fi Assist** off.

Checking a device is ready

1. Connect it to `pibo` or `pibo-24`.
2. Open a robot's address in the browser and leave the page open for two or three minutes.
3. Still connected — the device is ready. Moved to another network — one auto-join setting was missed.

Do this once, before handing the devices over

Preparing ten tablets on a table is far quicker than working out mid-lesson why one child's screen keeps going blank. Put it on the installation checklist.

5 Checking a robot

What to confirm before handing a robot over.

Check	Expected
Chest screen	SN, an I P address, and <code>pibo</code> on the AP line.
Block editor	Opening the IP address in a browser shows the editor, in English.
A test program	Open the <i>Basic</i> example and run it. A line appears in the Terminal.
Sound	Open the <i>Speech</i> example and run it. The robot speaks.
Movement	Open the <i>Motion</i> example and run it, with space around the robot.
Camera	Open the <i>Face</i> example and run it. A picture appears in the IDE.

What runs on the robot

Everything is served by the robot itself. Replace `<ip>` with the address on the chest screen.

Address	What it is
<code>http://<ip></code>	Block editor (IDE). The normal way in.
<code>http://<ip>:8080</code>	Reference documentation, also reached from the Guide button.
<code>http://<ip>:50000</code>	Tools — motion editor and sound recorder.
<code>http://<ip>:50010</code>	Classifier — image model training.
<code>http://<ip>:50020</code>	Language model server.

Open Tools, Classifier and the language model from the buttons in the editor rather than typing the addresses — only one of the three may run at a time, and the editor handles the switching.

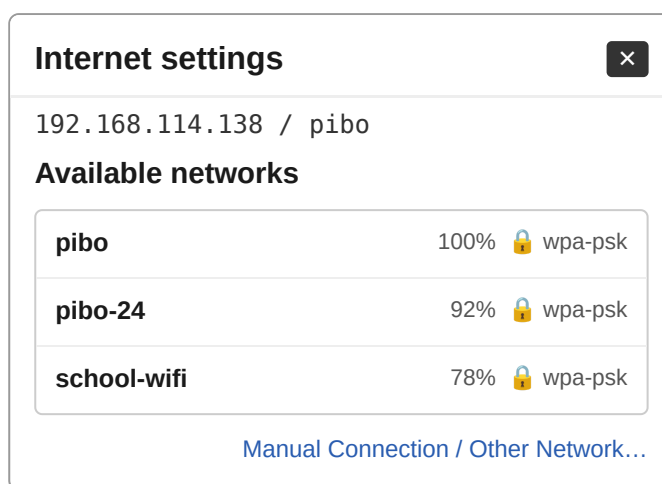
The documentation on the robot is in Korean

The Guide button on port 8080 still serves the Korean reference text in this software version. Use the printed *Teacher's Guide* instead. An English version is planned for a later release.

6 Connecting to a different network

Only needed if a robot has to join a network other than the supplied router — for example a school network, or after a factory reset when the router is not available. It has to be done on each robot separately.

1. Switch the robot on and wait about a minute. With no `pibo` network present it starts its own.
2. On a tablet or laptop, join the Wi-Fi network `pibo-` followed by the **SN** shown on the chest screen. The password is `!pibo0314`.
3. Open `http://192.168.34.1` in a browser.
4. Open **Internet settings**, choose the network from the list, enter its password and confirm.
5. The robot restarts by itself. After about a minute the chest screen shows an address on the new network.
6. Reconnect the tablet to that network and open the address to check.



Internet settings, reached from the block editor

If the network you want is not in the list

The list only shows networks the robot can actually use. A 5 GHz network on a high channel, or one using WPA3, will not appear. Networks whose names contain spaces are handled correctly. **Manual Connection** at the bottom covers hidden networks and WPA-Enterprise, which needs an identity as well as a password.

Confirming restarts the robot

Make sure nobody is in the middle of a lesson on that robot.

Going back to the supplied router

Set the Wi-Fi again by the same steps, choosing `pibo`. A factory reset (section 6) also puts the setting back to `pibo`.

7 Factory reset

Factory reset is in the block editor. It is useful when handing a robot to a new class, but it does more than most people expect and cannot be undone.

What it does

Action	Effect
Deletes saved programs	Everything children have saved on that robot is gone.
Deletes recordings, images and trained models	Custom motions, recorded sounds and any Classifier model are removed.
Restores the example programs	The ten built-in examples are put back as new.
Resets the Wi-Fi setting	Back to <code>pibo</code> .
Switches the robot off	It does not restart by itself.

With the supplied router this is harmless

The Wi-Fi setting goes back to `pibo`, which is what the supplied router provides — so the robot rejoins by itself when switched back on. Nothing needs to be set again.

But not if the robot was moved to another network

If you pointed the robot at a school network using section 6, a factory reset throws that setting away. The robot will come back in AP mode and section 6 has to be repeated. Tell teachers not to use factory reset on their own in that case.

Before resetting

1. Check nothing on the robot is worth keeping — saved programs, recorded motions, trained models.
2. Note which network the robot should be on afterwards.
3. Reset, then switch the robot back on and confirm the chest screen shows an address.

8 Troubleshooting

Symptom	Cause and remedy
Chest screen shows no IP address	The robot did not find <code>pibo</code> and is in AP mode. Check the router is switched on and broadcasting. If it is, move the robot closer and restart it. If one robot alone behaves this way while the others are fine, its Wi-Fi setting may have been changed — set it again using section 6.
All the robots went offline at once	Almost always the router. Check it has power, and that the 5 GHz channel is still fixed at 36 and the mode still 11ac — a firmware update or a factory reset of the router puts these back to defaults. Restore the settings in Appendix A, then restart the robots.
Some robots connect, others do not, in the same room	Usually too many robots on one router, or one of them is out of range. Count how many are on the router — ten is the recommended limit — and check the distance and any metal furniture in the way.
Robot connects, then drops repeatedly	First check whether a second router broadcasting <code>pibo</code> is within range — a spare one left switched on in the same room, or one in an adjoining room. The robot will move between the two and drop each time. Switch the extra one off. Otherwise the signal where the robot sits is too weak.
Connected, but the editor will not open	Check the tablet is on the same network as the robot, and that the address matches the chest screen. Then force a reload with Ctrl + Shift + R .
Camera or motion blocks do nothing	Tools or Classifier is open somewhere and is holding the hardware. Close those tabs. Running any program from the editor also releases them.
Pictures from the camera are slow	Several robots streaming video at once on a crowded router. Reduce the number of robots per router, or have half the class work on non-camera examples.
Robot will not start	Check the power adapter. The eyes light and a greeting plays within about twenty seconds of power being applied; if nothing happens at all, the robot needs service.

What to send when reporting a problem

The **SN** from the chest screen, the software version, what the chest screen shows, how many robots are on that router, and whether the problem affects one robot or all of them. That is usually enough to answer without a site visit.

Appendix A Defaults and addresses

Router settings — restore these if the router is reset

5 GHz network name	pibo
2.4 GHz network name	pibo-24
Password	!pibo0314
5 GHz channel	36, fixed
2.4 GHz channel	1, fixed
Channel width	80 MHz
Wireless mode	11ac — Wi-Fi 6 / 11ax off
Security	WPA2-PSK (AES) — not WPA3 or mixed
Band steering	Off
DHCP pool	100 addresses or more
Automatic firmware update	Off

Keep a settings backup file from the router's own backup / restore page. Restoring it is far quicker than re-entering everything.

Wi-Fi as the robots are shipped

Network the robot looks for	pibo (5 GHz)
Password	!pibo0314

AP mode

Network name	pibo- + the SN from the chest screen
Password	!pibo0314
Robot's address	192.168.34.1
Addresses given out	192.168.34.10 – 192.168.34.50
Band	2.4 GHz, so older laptops and USB adapters can connect

Ports on the robot

80	Block editor
8080	Reference documentation
50000	Tools
50010	Classifier
50020	Language model

Capacity

Routers per room	1 — never two with the robot network
Robots per router	10 recommended
More than 10 robots	Split between rooms, one router each, far enough apart that the networks do not overlap

Passwords are the same on every robot and router

The values above are shipped identically on all units. Anyone who knows them can join. Decide with the school whether to keep them or set your own, and record whatever you choose. Changing the Wi-Fi password means setting every robot again by section 6, so decide before installation rather than after.